

ARTICLE



STEM CELL TRANSPLANTATION

Disturbances in microbial skin recolonization and cutaneous immune response following allogeneic stem cell transfer

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The composition of the gut microbiome influences the clinical course after allogeneic hematopoietic stem cell transplantation (HSCT), but little is known about the relevance of skin microorganisms. In a single-center, observational study, we recruited a cohort of 50 patients before undergoing conditioning treatment and took both stool and skin samples up to one year after HSCT. We could confirm intestinal dysbiosis following HSCT and report that the skin microbiome is likewise perturbed in HSCT-recipients. Overall bacterial colonization of the skin was decreased after conditioning. Particularly patients that developed acute skin graft-versus-host disease (aGVHD) presented with an overabundance of *Staphylococcus* spp. In addition, a loss in alpha diversity was indicative of aGVHD development already before disease onset and correlated with disease severity. Further, co-localization of CD45⁺ leukocytes and staphylococci was observed in the skin of aGVHD patients even before disease development and paralleled with upregulated genes required for antigen-presentation in mononuclear phagocytes. Overall, our data reveal disturbances of the skin microbiome as well as cutaneous immune response in HSCT recipients with changes associated with cutaneous aGVHD.

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INTRODUCTION

A primary role of the skin is to serve as a physical barrier against invading pathogens [1, 2], whilst simultaneously hosting various microorganisms, including bacteria, fungi and viruses that compose the skin microbiome [3]. In allogeneic hematopoietic stem cell transplantation (allo-HSCT), however, skin integrity can become impaired due to conditioning, leading to translocation of microbes from the stratum corneum into deeper tissue compartments [4]. Acute graft-versus-host disease (aGVHD) represents the most frequent complication in HSCT and is defined by allo-activation of donor T cells by host tissues [5] and residual host immune cells [6], leading to inflammation of the skin, liver and gut [7].

Recent studies have emphasized the role of intestinal microbes in GVHD development [8] and expansion of *Enterococcus* and loss of *Clostridiales* have been observed to correlate with transplant-related mortality [9, 10]. However, the impact of the skin microbiome in GVHD remains elusive. Nevertheless, skin microbiota have been

shown to play a role in many inflammatory skin conditions, most prominently atopic dermatitis. Atopic lesions are often dominated by *Staphylococcus* spp. [11], resulting in the development of the first bacteriotherapy successfully tested in a clinical trial [12].

In the current study, we investigated the role of both the skin and intestinal microbiome, correlating changes in microbiota with the cutaneous immune response and clinical course up to one year after stem cell transfer.

METHODS

Sampling

Adults (>18years; $n = 50$) previously diagnosed with a hematologic malignancy and suitable allogeneic HSCT were recruited before conditioning at the Medical University of Vienna after ethical approval (ECS 1087/2016) and participated voluntarily and with fully informed written consent. Skin scrapes from different regions (palm, back and face) and stool were obtained before conditioning ($n = 180$), the day of transplantation (Tx, $n = 171$), at day

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14 ($n = 174$), day 100 ($n = 130$) and day 365 ($n = 109$). Additional samples were collected from healthy, age-matched volunteers ($n = 15$) and empty tubes were used as negative controls ($n = 71$). 6 mm skin biopsies from a subset of patients ($n = 9$) were obtained at the same 5 time points, while additional biopsies were collected after aGVHD onset ($n = 12$). aGVHD lesions and unaffected skin were also swabbed for isolation of *Staphylococcus* spp. ($n = 5$). All samples from GVHD patients were collected before start of any treatment (Supplementary Table 13). GVHD was diagnosed according to the Glucksberg criteria for acute [13] and the NIH consensus criteria for chronic GVHD [14]. All patients were administered prophylactic antibiotics (Supplementary Fig. 3G) and other antimicrobial agents upon hospitalization (Acyclovir), during their neutropenic period until cell engraftment (Paromomycin, Metronidazol and Nystatin) as well as upon development of neutropenic fever (Piperacillin/Tazobactam).

16S rRNA gene amplicon sequencing

DNA was extracted using the QIAamp DNA Microbiome Kit with additional lysis steps using Mutanolysin (0.01 mg/ml), Lysozyme (1 mg/ml) and Lysostaphin (0.1 mg/ml). Primer pairs 341 F/785 R [15] were used in a two-step PCR workflow [16]. Amplicon sequence variants (ASVs) were inferred using the DADA2 R package [17, 18]. After trimming, sequences were classified with SINA version 1.6.1 [19] and SILVA database SSU Ref NR 99 release 138 [20]. Eukaryota, mitochondria, chloroplast, and known PCR contaminations (ASV_qdi_vob) were removed. Alpha diversity was calculated using the R ampvis2 v.2.6.5 [21] and Bray Curtis using the R package 2.5–6 [22].

For differential abundance analysis only, skin/stool samples containing between 900/2000 and 27000/60000 reads were used from patients with a minimum of three skin and two stool samples. Additionally, samples with a GVHD after d100 were excluded. For pairwise differential abundance testing DESeq2 was run with default parameters except fitType = 'local'.

Immunofluorescence

Acetone-fixed cryosections from skin were hybridized with a 16S rRNA-probe (EUB338: 5'-GCTGCTCCCGTAGGAGT-3' [23]) as well as *Staphylococcus*-specific probe (STAPHY2: 5'-TCCTCCATATCTCTGCGC-3' [24]). Reverse complements were used as negative controls. Hybridization was performed in 0.9 M NaCl/20 mM Tris-HCl/0.01% SDS for 2 h at 46 °C (EUB338) and 49 °C (STAPHY2). After washing at 48 °C (EUB338) and 46 °C (STAPHY2) for 15 min in 0.215 M NaCl/20 mM Tris-HCl/0.01% SDS, samples were stained with CD45 antibodies and DAPI. Images were acquired using the TissueFAXS software (V6.01.126) and bacteria were detected with StrataQuest (V6.01.167; TissueGnostics GmbH). Epidermis and dermis (<500 µm) were analyzed separately, while blood vessels and hair follicles were excluded.

Co-localization score

Co-localization of bacteria and CD45⁺ cells was calculated with an adapted version of the spatial arrangement analysis-function from DAIME [25]. As our input data contained the centroid position of cells, we adjusted the formula by calculating the total number of hits at range r as the cumulative hits at or below r and $r-1$ ($H(r) = H_cumm(r) - H_cumm(r-1)$). Hits or misses are replaced by the product of possible dipoles at range r (the circumference of a circle of radius r : $2 * \pi * r$) and the total area (At) ($T(r) = 2 * \pi * r * At$). In addition, population densities ($D1$ and $D2$) are calculated using the number instead of the area of one population ($N1$ or $N2$) through the total image area (At): $D1 = N1/At$, $D2 = N2/At$).

$$g(r) = \frac{[H_cumm(r) - H_cumm(r-1)] * At}{[(2 * \pi * r) * (2 * N1 * N2)]}$$

Fluorescence-activated cell sorting

Cutaneous single-cell suspensions were stained for surface markers after collagenase (0.5 U/ml) and DNase I (10 mg/ml) digestion [6]. All antibodies are indicated in Supplementary Table 17. For Smart-seq2 bulk-RNA sequencing 100 cells of both T cells (CD45⁺CD3⁺HLA-DR) and mononuclear phagocytes (CD45⁺CD3⁺HLA-DR^{high}), were sorted into lysis buffer containing RNase inhibitor (1.25 U/ml). All stainings were adapted to isotype-matched controls and gated accordingly.

Smart-seq2 bulk RNA sequencing

cDNA synthesis was performed using the Smart-seq2 protocol [26] and sequenced on an Illumina HiSeq 3000/4000 platform at the 50-bp (base

pair) single-read configuration. Read trimming was conducted using ReadTools (v.1.0.0) [27] and mapped using STAR (v2.5.3a) [28]. Reads mapped in multiple genomic locations were excluded by SAMtools (v1.4) [29]. Read counts were generated using the featureCounts function from the Subread package (v.1.22.1) [30]. Gene expression analysis was done with DESeq2 (v.3.22.3) [31]. Normalized count (log2) by DESeq2 was used for t-SNE and heatmaps. Pathway enrichment analysis was done by using the KEGGA and GOANA methods, respectively, from the limma package [32].

Co-cultures

Keratinocytes isolated from skin removed in abdominoplasty surgeries using Dispase II (4 U/ml) and PBMCs obtained by Ficoll-Paque reagent were co-cultivated with heat-inactivated *S. epidermidis* from three aGVHD patients at MOI 0.1, 1 and 10 for 8 h. *Staphylococci* were grown on a Columbia CNA agar with 5% sheep blood at 37 °C and 10% CO₂. Identification to species level was performed by MALDI-TOF MS. From keratinocytes, RNA was isolated using the RNeasy Mini Kit and RT-qPCR was performed: 10 min at 55 °C, 1 min at 95 °C, 40 cycles of 10 sec at 95 °C and 1 min at 60 °C. All primers are listed in Supplementary Table 18. Stimulated PBMCs were labeled using surface antibodies followed by intracellular cytokine staining (see Supplementary Table 17) [33].

Whole-genome sequencing

DNA was extracted using the DNeasy PowerSoil Pro Kit, followed by library preparation (NEBNext[®] Ultra[™] II FS DNA Library Prep Kit) and sequencing on the Illumina MiSeq platform (v3 chemistry, 600 cycles). Sequencing reads were quality trimmed using cutadapt v.2.10 [34] and assembled by SPAdes v.3.14.1 [35]. Contigs shorter than 1 kbp were removed with seqtk v.1.3 (github.com/lh3/seqtk). Assembly quality was assessed with CheckM v.1.1.1 [36] and genomes were compared to each other using fastANI v.1.3 [37]. Genomes were annotated using prokka v.1.14.5 [38] and protein orthologs were detected using proteinortho v.6.0.24 [39].

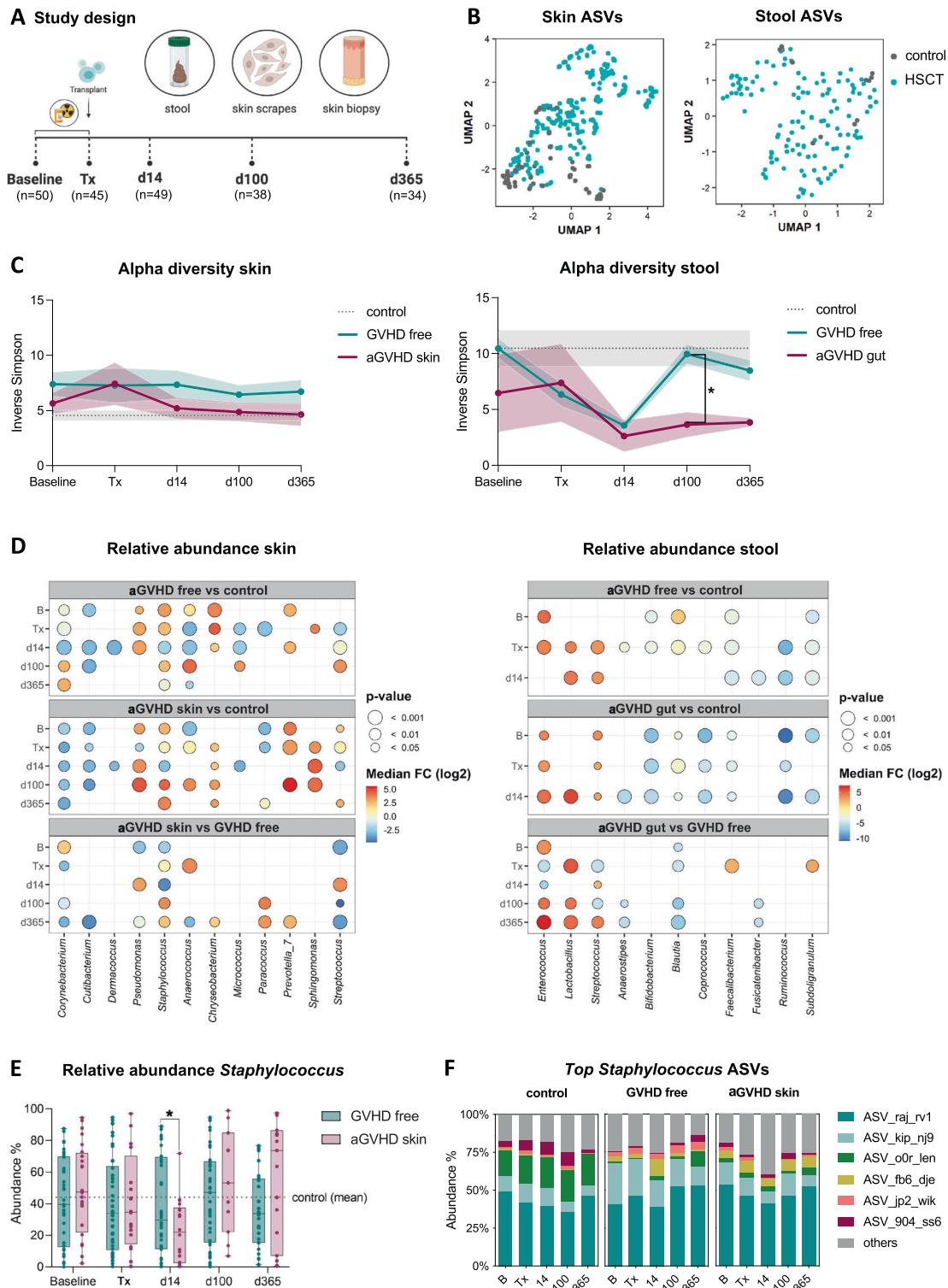
Statistical analysis

For normally-distributed data, one-way ANOVA and Bonferroni pairwise comparisons were used, while for non-normally distributed data, Kruskal-Wallis and Dunn's post hoc tests were applied. Competing risk analysis for GVHD with the competing events death and relapse were calculated using the R-package cmprsk. For patients with GVHD, covariates were compared between mild and severe GVHD-patients by t-, Mann-Whitney- and Chi-squared tests. Mixed linear models were analyzed including all covariates with random factor patient ID in the full patient set. Changes of mean alpha diversity in skin/stool were analyzed by generalized estimating equations. P values smaller than 0.05 were considered significant.

RESULTS

HSCT disrupts the skin and gut microbiome

The effect of allo-HSCT on the cutaneous and gut microbiome as well as skin immunity was assessed in 50 patients up to one year after transplantation (Fig. 1A, Supplementary Tables 1–4). Compared to healthy volunteers ($n = 15$), we identified HSCT-specific clustering, particularly in the skin (Fig. 1B, Supplementary Fig. 1A–E). While both acute (skin $n = 10$, gut $n = 6$) and chronic GVHD (skin $n = 4$, gut = 0) cases were recorded during the one-year observation period, only acute GVHD (aGVHD) patients were considered for this analysis. Alpha diversity of the gut microbiome was disrupted early after HSCT (day of transplantation until d14), which was associated with aGVHD diagnosed by Glucksberg criteria [13] (mean [CI] d100 aGVHD free 3.65 [0.55, 6.74]; aGVHD 9.96 [8.17, 11.74]; $p = 0.0338$; Fig. 1C). However, Bray Curtis beta diversity showed a long-lasting shift of both the skin and intestinal microbiome compared to controls 14 days after transplantation (mean [CI] skin control 0.4 [0.28, 0.52] vs. skin HSCT 0.63 [0.57, 0.7]; $p = 0.0016$ and gut control 0.54 [0.37, 0.66] vs. gut HSCT 0.9 [0.8, 1.0]; $p = 0.011$; Supplementary Fig. 3A, B). While alpha diversity measures microbial diversity within a single sample, beta diversity estimates the similarity/dissimilarity in-between samples.



We next investigated changes in relative abundance of bacteria after HSCT. In skin, commensals such as *Corynebacterium*, *Cutibacterium* and *Dermacoccus* were decreased (Fig. 1D, Supplementary Fig. 2A), while *Pseudomonas* were increased 14 days after transplant in HSCT recipients (mean [CI] baseline 0.7% [0.37, 1.01]; d14 2.12% [1.12; 3.15]; $p = 0.012$; Supplementary Fig. 3C). Further, *Staphylococcus* spp. showed first a lower abundance at d14 (mean [CI] control

50.66% [37.19, 64.13]; skin aGVHD 24.58 [14.02, 35.13]; $p = 0.034$) and later an expansion at d100 and d365 in skin aGVHD patients (Fig. 1D, E, Supplementary Fig. 2A). Overabundance of staphylococci was not due to a particular ASV (amplicon sequencing variant, Fig. 1F). In stool, loss of butyrate-producing bacteria after conditioning was paralleled by an expansion of *Enterococcus*, *Lactobacillus* and *Streptococcus* (Fig. 1D, Supplementary Figs. 2, 3D–F).

Fig. 1 Conditioning regimen induces changes in stool and skin microbiome. **A** Study timeline and sampling strategy of patient cohort; $n = 50$; Tx = day of transplantation. **B** Unsupervised clustering of amplicon sequence variants (ASVs) from skin and stool samples obtained from transplanted patients (blue dots) and healthy controls (grey dots). Each dot represents one sample and time point. **C** Alpha diversity (Inverse Simpson) of skin/stool samples from control (dotted line), GVHD free (turquoise line) and patients that develop skin/gut aGVHD (purple line). For control samples an average in alpha diversity over all time points was calculated. After testing for normal distribution (Shapiro-Wilk test), a non-parametric test (Kruskal-Wallis test and Dunn's multiple test correction) was performed comparing groups at specific time points. Mean $\pm$ SEM, $*P < 0.05$. **D** Median log2-fold-changes $p < 0.05$ of skin/stool bacteria (genus level) in control vs. GVHD free (top panel), control vs. aGVHD skin/gut (middle panel) and GVHD free vs. aGVHD skin/gut patients (bottom panel). **E** *Staphylococcus* abundance (%) of skin samples. Statistical testing from relative percentages conducted as described in **C**. **F** Abundance of top 6 *Staphylococcus* ASV as a proportion of total *Staphylococcus* read counts (%). ASV_raj_rv1: *S. aureus/capitis/caprae/epidermidis/haemolyticus/warneri*, ASV_kip_nj9: *S. epidermidis/haemolyticus/hominis/petrassii*, ASV_o0r_len: *S. aureus/capitis/caprae/epidermidis/haemolyticus/saprophyticus/warneri*, ASV_fb6_dje: *S. aureus/devriesei/epidermidis/haemolyticus/hominis*, ASV_jp2_wik: *S. epidermidis*, ASV_904_ss6: *S. aureus/pasteuri/warneri* and Others: less abundant *Staphylococcus* ASVs. Time-wise comparisons of ASVs in control-, GVHD free- and aGVHD skin patients were calculated using Kruskal-Wallis test and Dunn's multiple test correction.

Low alpha diversity and high staphylococci in skin aGVHD

We next investigated correlations between clinical outcomes and microbial signatures, however, only acute GVHD cases were included in the analysis, except for statistical models (Supplementary Tables 5–12). Alpha diversity in skin was lower after cutaneous aGVHD onset (mean [CI] before onset 7.46 [5.56, 9.28]; after onset 3.84 [2.71, 4.97]; $p = 0.0242$), while no differences were observed in the gut, although this comparison was highly underpowered given the low number of intestinal GVHD patients ($n = 6$, Fig. 2A). Generalized estimated equations (GEE) confirmed these results, while also showing a significant decrease in diversity in stool samples (estimate [CI] skin -4.16 [-6.811 , -1.287]; $p = 0.006$ and gut -5.1901 [-7.17 , -3.21]; $p < 0.0001$; Supplementary Tables 5, 6). As all patients were administered antibiotics prior and during the transplantation (Supplementary Fig. 3G), we next investigated the effect of different antimicrobial agents on alpha diversity. GEE revealed a negative effect of Piperacillin/Tazobactam on diversity in skin GVHD (estimate [CI] -4.93 [-8.684 , -1.182]; $p = 0.001$; Supplementary Table 7), while no effect was observed for stool samples. In addition, competing risk analysis showed no cofounders for GVHD development, including variables such as age, gender, diagnosis prior to HSCT, conditioning regimen as well as number and type of antibiotics (Supplementary Tables 8–11).

In skin, the degree of alpha diversity reduction before disease onset was predictive of disease severity (mean [CI] mild skin aGVHD 8.66 [6.33, 10.98], severe skin aGVHD 3.93 [2.44, 5.42], $p = 0.039$; Fig. 2B). We again observed an increased relative abundance of staphylococci after aGVHD onset (mean [CI] before onset 39.5% [32.4, 46.59]; after onset 57.06% [43.69, 70.42]; $p = 0.02$; Fig. 2C, Supplementary Table 12), which was not due to one particular ASV (Fig. 2D). While a significant increase in staphylococci abundance was observed after onset of mild skin aGVHD (mean [CI] before onset 39.2% [30.66, 47.8]; after onset 58.52% [44.17, 72.87]; $p = 0.0431$), no difference was observed in severe cases (Fig. 2E).

Functional evaluation of *S. epidermidis* in skin aGVHD

To address our findings of staphylococcal overgrowth in skin aGVHD, we performed FISH targeting bacteria (EUB338) and only *Staphylococcus* spp. (Staphy2) in aGVHD lesions ($n = 12$). Together with a high infiltration of CD45⁺ cells (Supplementary Fig. 4A), colonization with staphylococci was increased in the epidermis of lesional skin compared to unaffected controls (mean [CI] aGVHD free 12.04 [9.05, 15.03]; skin aGVHD 54.96 [41.47, 68.45]; $p = 0.012$; Fig. 3A, Supplementary Fig. 4B).

Based on these observations, we next performed whole-genome sequencing of *Staphylococcus* spp. isolates from untreated aGVHD lesions and unaffected skin sites. Cultivation of skin swabs prior to sequencing confirmed increased growth of staphylococci in lesional skin (Fig. 3B). Interestingly, *S. aureus* only

grew in one out of 6 lesions (Supplementary Table 14), while *S. epidermidis* was the only species found in all samples. We therefore performed sequencing of 6 lesional and 6 non-lesional *S. epidermidis* strains (Supplementary Tables 15, 16) and found an average nucleotide identity of 99% across all samples except one, with a 97% sequence identity (Supplementary Fig. 4C). Around 75% of genes were shared between all isolates (1983 core vs. 717 flexible genes), while most differences were found in-between patients (Fig. 3C). Only few genes were unique to lesional (36) and non-lesional skin sites (35). Co-cultivation experiments of the same *S. epidermidis* strains with primary human keratinocytes and peripheral blood mononuclear cells (CD14⁺ monocytes and CD3⁺ T cells) confirmed this observation, showing no differences in regulation of antimicrobial peptides and inflammatory cytokines between lesional- and non-lesional isolates (Fig. 3D, E, Supplementary Fig. 4D).

Skin recolonization in HSCT

To investigate bacterial colonization and the cutaneous immune response following HSCT, we combined rRNA-targeted FISH (EUB338) with antibodies for CD45 (Fig. 4A). Interestingly, not only CD45⁺ cells (Fig. 4B), but also bacteria were decreased after conditioning in the epidermis (mean [CI] EUB338 baseline 35.25 [27.71, 42.78] vs. d14 15.51 [11.03, 19.9]; $p = 0.0005$; Fig. 4C). A similar reduction was observed for staphylococci by using a FISH-probe targeting only *Staphylococcus* spp. (mean [CI] Staphy2 baseline 22.56 [18.39, 26.73] vs. d14 5.37 [3.47, 7.27]; $p = 0.005$; Fig. 4D). However, numbers of CD45⁺ cells, total bacteria and staphylococci were independent of skin aGVHD development (Supplementary Fig. 5A–C).

Further, spatial arrangement analysis adapted from the DAIME software [25] (Supplementary Fig. 6A–C) revealed closer proximity of bacteria and CD45⁺ cells 14 days after transplantation in the epidermis of patients that later developed aGVHD (mean [CI] GVHD free 0.09 [-0.02 , 0.19]; aGVHD 1.5 [0.18, 2.83]; $p = 0.0211$, Fig. 4E left), even though both bacteria and leukocytes were reduced in overall numbers at that time (Fig. 4B, C). A similar co-localization in the epidermis, but not in the dermis, was observed for *Staphylococcus* spp. prior to aGVHD onset (mean [CI] GVHD free 0.11 [0.02, 0.19]; aGVHD 3.01 [-0.41 , 6.55]; $p = 0.0341$; Fig. 4E right, Supplementary Fig. 5D).

Activation of host immunity after conditioning

To assess the consequences of microbial disruption on skin immunity, we evaluated the transcriptome of isolated skin T cells (CD45⁺CD3⁺HLA-DR⁺) and mononuclear phagocytes (MNP, CD45⁺CD3⁺HLA-DR^{high}) at the 5 study timepoints by bulk-RNA seq (Fig. 5A). Upon unsupervised clustering, transcriptomes of MNPs and T cells were clearly distinguishable, while no time-specific clustering was observed (Supplementary Fig. 7A, B). The highest number of differentially regulated genes were observed

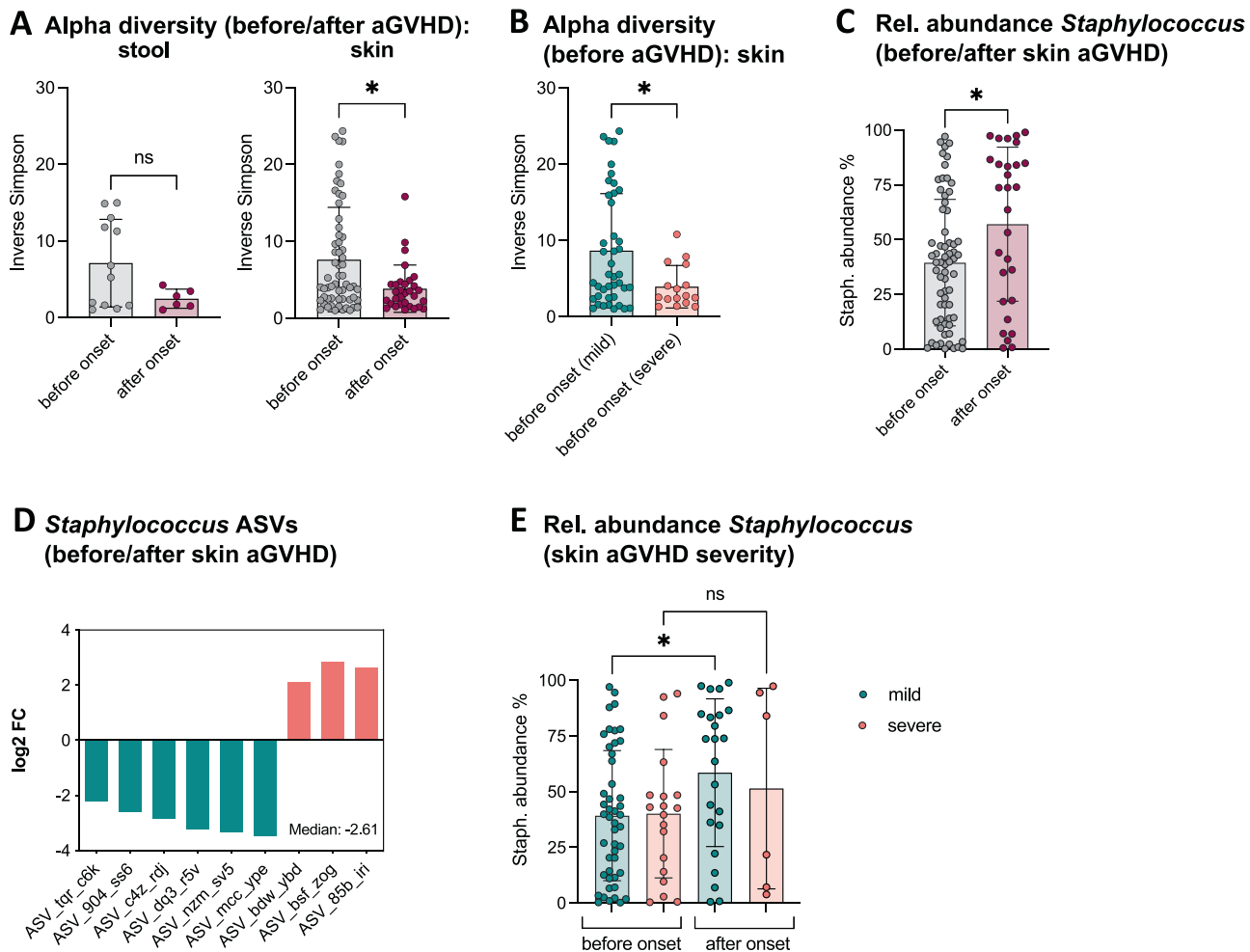


Fig. 2 Low alpha diversity and *Staphylococcus* overgrowth in skin aGVHD. **A** Alpha diversity (Inverse Simpson) of stool and skin samples before (grey) and after aGVHD onset (purple). Skin: before onset $n = 14$, after onset $n = 13$ and stool: before onset $n = 5$, after onset $n = 4$. For skin/stool samples data from different timepoints were pooled. **B** Skin samples before aGVHD onset stratified by severity (mild: grade 1–2 and severe: grade 3–4; $n = 9$ and $n = 5$, respectively). After testing for normal distribution (Shapiro-Wilk test), a non-parametric test (Mann-Whitney test) was performed. Mean $\pm$ SEM, $*P < 0.05$. **C** *Staphylococcus* abundance in % and **D** log2 fold changes of *Staphylococcus* ASVs before vs. after aGVHD onset; ASV_tqr_c6k, ASV_mcc_ype & ASV_c4z_rdj: *S. epidermidis*, ASV_904_ss6: *S. aureus/pasteuri/warneri*, ASV_dq3_r5v & ASV_nzm_sv5: species unknown, ASV_bdw_ybd: *S. aureus/haemolyticus/hominis/petrasii*, ASV_85b_iri: *S. hominis* and ASV_bsf_zog: *S. saccharolyticus*. **E** *Staphylococcus* abundance in % by aGVHD severity. Kruskal-Wallis test and Dunn's multiple test correction. Mean $\pm$ SD, $*P < 0.05$.

from Tx until d14 (MNP:728& 822; T cells: 448& 411, Fig. 5B), suggesting an early influence of conditioning on host immunity. In MNPs, we observed enrichment of cytosolic DNA-sensing and NOD-like receptor signaling pathways at d14, including *CASP4*, an essential regulator in NLRP3 inflammasome-dependent IL-1 release [40] and phago-lysosomal fusion genes *PSME1*, *P4HB* and *PRCP* [41, 42]. Additionally, antigen receptor-signaling was upregulated, including genes *PSMB3* and *PSMD4* as well as *TAPBP*, which facilitates peptide loading [43] (Fig. 5C, D). Further, enrichment of type 1 interferon signaling (Fig. 5D) was observed, which promotes dendritic cell activation [44, 45].

In parallel, T cells showed enriched responses to LPS and positive regulation of NF-kappaB signaling. In addition, *MYO1G*, a gene involved in T cell interaction with antigen-presenting cells [46] was increasingly expressed at d14. At d100, T cell receptor signaling was enriched together with granzymes (Fig. 5C, D). Assessment of receptor-ligand pairs [47] between T cells and MNPs confirmed increased TCR co-stimulation by upregulation of *CD28-CD80* and *CTLA4-CD80* at d100 and d365 (Fig. 5G).

Additionally, receptor-ligand pairs involved in chemotaxis and cell migration were increasingly expressed at d14, including *CCR7-CCL21*, *CXCR3-CXCL9*, *CCL2-CCR2* and *CCR1-CCL5* (Fig. 5G).

DISCUSSION

Thus far, studies of the relationship between microbiota and HSCT have focused on the intestinal tract, reporting a correlation between low diversity and risk of GVHD development [9, 10, 48]. These reports are corroborated by the successful use of fecal microbiota transplantation to treat intestinal GVHD [49, 50]. In the current study, we confirmed the loss of microbial diversity and expansion of *Enterococcus* and *Streptococcus in the gut* [9, 51], while butyrate-producing bacteria were reduced [52, 53].

In contrast, the skin microbiome remained more stable following HSCT, while colonization with bacteria was reduced after conditioning. Additionally, recolonization with *Staphylococcus* spp. was altered and an overgrowth of staphylococci in skin

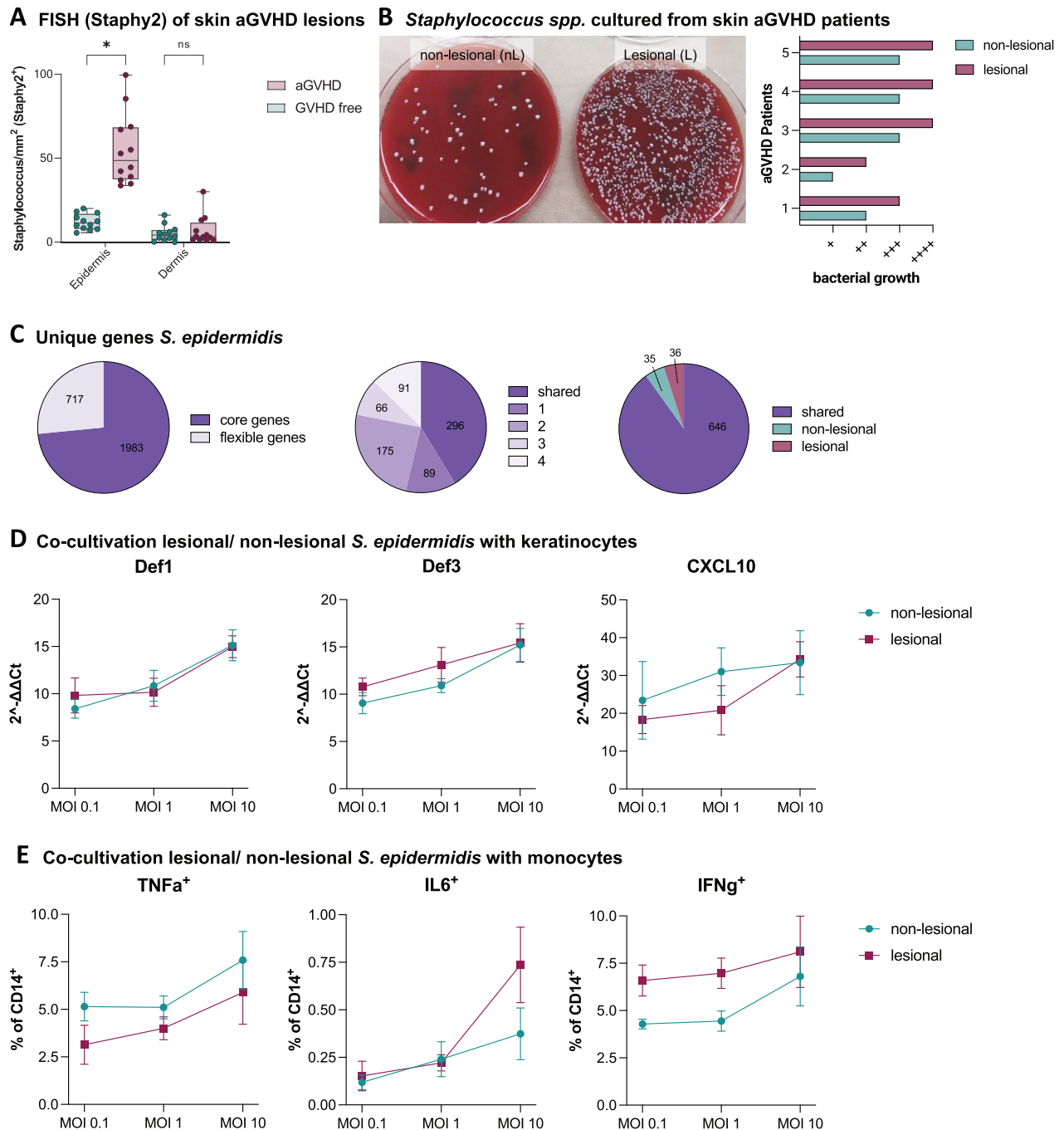


Fig. 3 Functional analysis of *Staphylococcus* spp. from lesional/ non-lesional skin. **A** Quantification of *Staphylococcus* using FISH (Staphy2) in acute GVHD lesions. GVHD free controls were time matched to lesional aGVHD samples ($n = 12$). Kruskal-Wallis and Dunn's post hoc tests. Mean $\pm$ SD. **B** Growth of *Staphylococcus* isolates from non lesional (nL) and lesional (L) skin on blood agar plate. Bacterial growth is indicated by +: minimal, ++: sparse, +++: moderate, and ++++: heavy ($n = 5$). **C** Pie charts showing unique and shared genes between all samples (top), individual patients (middle) and L vs. nL swabs (bottom). **D** Real-time qPCR and **E** FACS of human primary keratinocytes and blood CD14⁺ cells, respectively, co-cultivated with L/nL *S. epidermidis* isolated from 3 different patients for 8 h. Average and standard deviation of three biological replicates are depicted. After testing for normal distribution (Shapiro-Wilk test), a non-parametric test (Kruskal-Wallis test and Dunn's multiple test correction) was performed.

aGVHD was observed by 16S rRNA gene amplicon sequencing, FISH and cultivation. These findings are supported by Gu *et al.*, who also reported an increase in *Staphylococcus* spp. in skin GVHD lesions [54]. However, our data demonstrate that staphylococci are already perturbed before aGVHD onset, possibly a result of altered recolonization dynamics after conditioning.

Further, the cutaneous immune response after HSCT showed an activation of antigen-presenting cells together with increased T cell receptor-signaling and response to LPS, which are known pathomechanistic factors of aGVHD development [55]. In addition, spatial rearrangement analysis revealed an increased co-localization of staphylococci with CD45⁺ cells early after stem cell

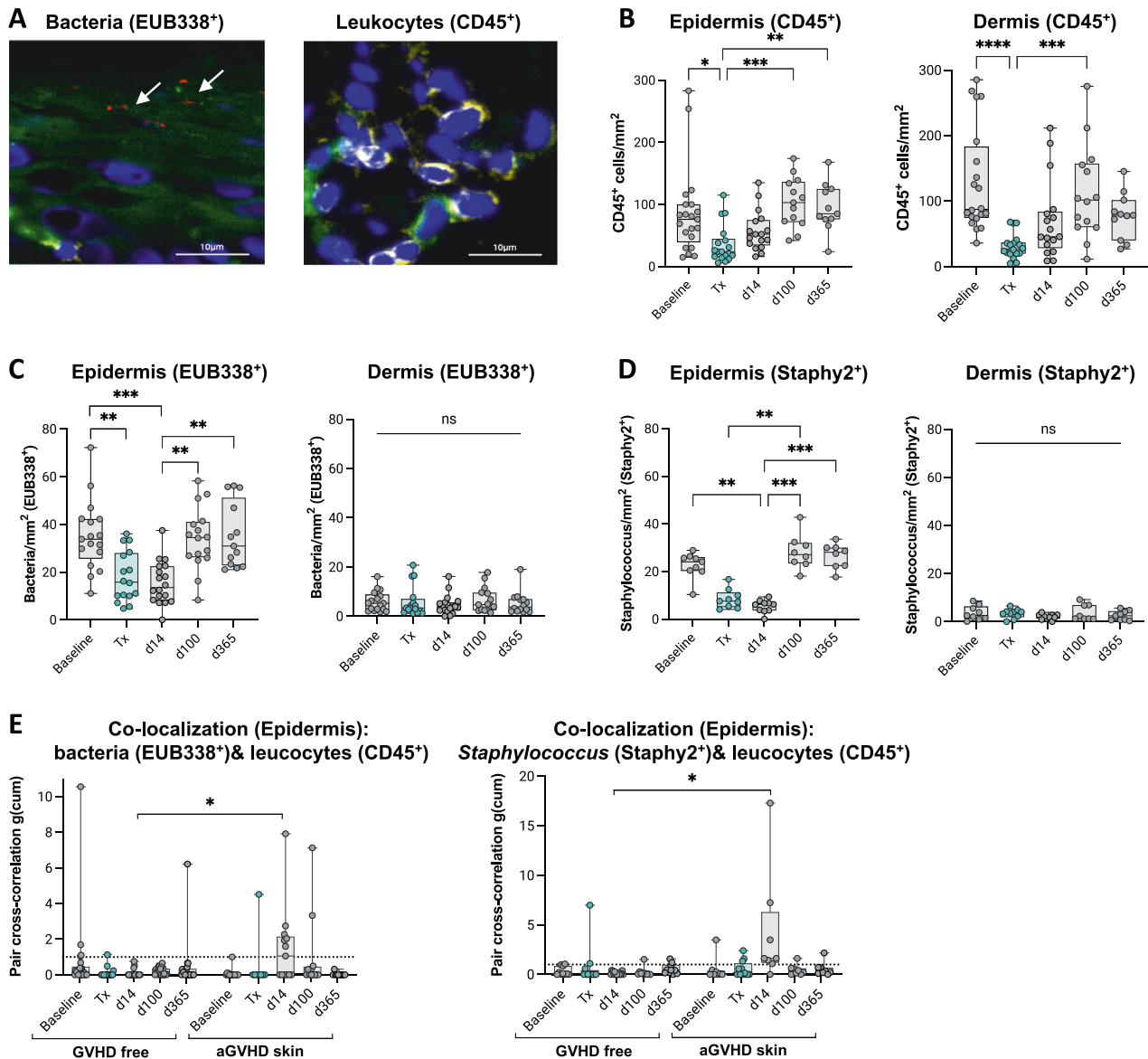


Fig. 4 Bacterial FISH reveals changes in skin colonization after conditioning. **A** Representative immunofluorescence image of bacteria labeled with a rRNA-targeted FISH probe (EUB338⁺ denoted by arrows; left) and CD45⁺ cells (right). Quantification of **B** CD45⁺ cells, **C** bacteria (EUB338⁺) and **D** staphylococci (Staphy2⁺) in both epidermis and dermis ($n = 19$). **E** Spatial analysis of bacteria (EUB338⁺; left) and Staphylococci (Staphy2⁺; right) with CD45⁺ cells at a radius $g(\text{cum})$ of 50 μm . Values of $g(\text{cum})$ above or below the dotted line indicate an attractive or repulsive relationship between the two populations. Normal distribution was tested using a Shapiro-Wilk test. Consequently, Kruskal-Wallis and Dunn's post hoc tests were applied. Mean $\pm$ SD, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

transfer, but only in patients that later developed skin aGVHD. Skin commensals have been shown to accelerate skin graft rejection in mice by increasing differentiation of alloreactive T cells [56]. Therefore, we suggest that the cutaneous immune system in aGVHD patients is increasingly reactive towards skin bacteria after conditioning, which correlates with previous findings of LPS-induced inflammation in gastrointestinal GVHD [57–60].

Although we didn't identify virulence factors of *Staphylococcus* spp. specific for aGVHD lesions in a small number of isolates as previously observed in Netherton syndrome [61], we provide evidence that overabundance of staphylococci is associated with skin GVHD lesions by both molecular (16S amplicon sequencing and FISH) and culture-based approaches. Whether also stain specific effects might have an impact on skin aGVHD development, as already described for intestinal GVHD [9, 51], needs to be

evaluated in a larger collection of *Staphylococcus* spp. isolated from untreated, skin aGVHD lesions. Other limitations of this study include that 16S RNA gene amplicon sequencing does not allow to distinguish between bacterial species. In addition, due to the prospective nature of this study, intestinal GVHD patients as well as chronic skin and intestinal GVHD patients were underrepresented in this study. Therefore, a distinction between patients with acute ($n = 10$) and chronic skin GVHD ($n = 4$) was not feasible.

In conclusion, our data suggest that changes in recolonization dynamics after HSCT and overgrowth of *Staphylococcus* spp. in skin aGVHD lesions may augment antigen-presentation in MNP and increased activation of T cells, which are well known hallmarks of GVHD pathogenesis [55]. Furthermore, our study sets the stage for projects focusing on communication of skin immune cells with the local microbiota and its involvement in cutaneous inflammation.

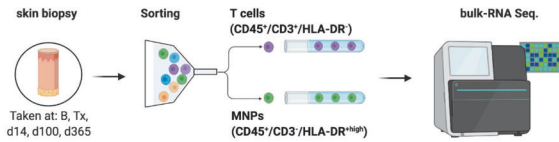
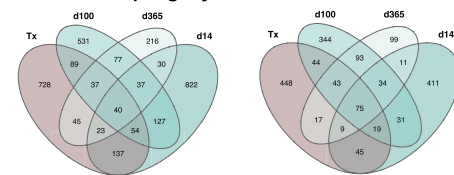
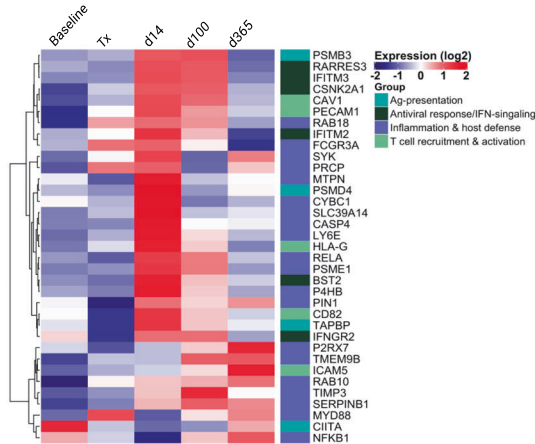
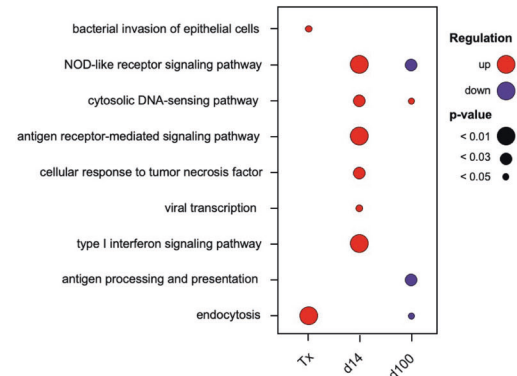
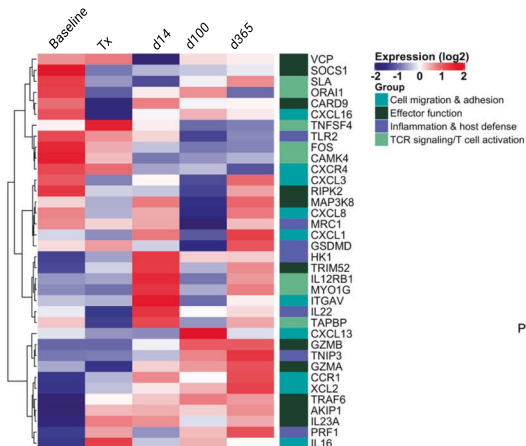
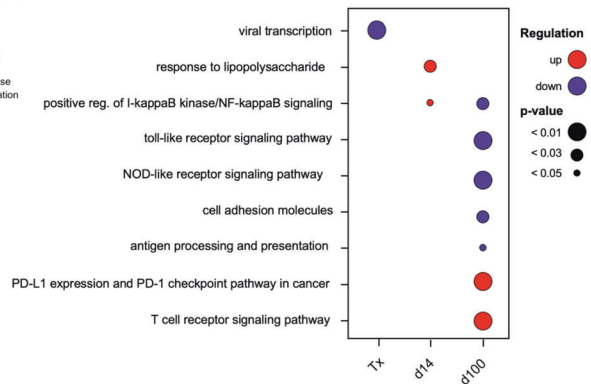
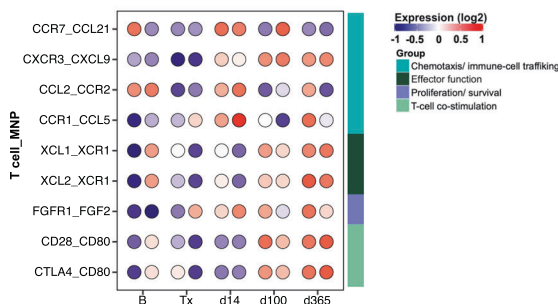
A Sample workflow**B Differentially regulated genes: mononuclear phagocytes****C DEGs: mononuclear phagocytes****D Pathway analysis: mononuclear phagocytes****E DEGs: T cells****F Pathway analysis: T cells****G Receptor-ligand pairs: T cells & mononuclear phagocytes**

Fig. 5 Activation of host immunity via conditioning therapy. **A** Visualizations of sample processing of skin biopsies taken at 5 study time points for isolation of T cells (CD45⁺CD3⁺HLA-DR⁻, $n = 9$) and mononuclear phagocytes (MNPs, CD45⁺CD3⁺HLA-DR^{high}, $n = 6$) for bulk-RNA seq; Baseline (B), day of transplantation (Tx). **B** Venn diagram of isolated skin MNPs and T cells indicating unique and shared differentially expressed genes (DEGs) for each time point versus baseline. Log2-fold-changes ($p < 0.05$) of selected genes in cutaneous **C** MNPs and **E** T cells related to immune function. Enriched, immune-related GO- and KEGG-pathways for cutaneous **D** MNPs and **F** T cells at Tx, d14 and d100 versus baseline samples. **G** Gene signatures of receptor-ligand pairs (from cellphonedb) in T cells and MNPs during HSCT procedure.

DATA AVAILABILITY

16S rRNA gene amplicon and whole-genome sequencing data are available under the BioProject accession number PRJNA804670. Bulk RNA sequencing data are deposited in NCBI's Gene Expression Omnibus database (GSE198625).

CODE AVAILABILITY

The code for the co-localization score is available via <https://github.com/ramvinay/HostMicrobiome>.

REFERENCES

- Gallo RL. Human Skin Is the Largest Epithelial Surface for Interaction with Microbes. *The J Investigative Dermatol*. 2017;137:1213–4.
- Grice EA, Segre JA. The skin microbiome. *Nat Rev Microbiol*. 2011;9:244–53.
- Kabashima K, Honda T, Ginhoux F, Egawa G. The immunological anatomy of the skin. *Nat Rev Immunol*. 2019;19:19–30.
- Brown VI. HSCT-Associated Complications of the Skin, Hair, and Nails. In: Brown VI, editor. *Hematopoietic Stem Cell Transplantation for the Pediatric Hematologist/Oncologist*. Cham: Springer International Publishing; 2018. p. 363–8.
- Zeiser R, Blazar BR. Acute Graft-versus-Host Disease - Biologic Process, Prevention, and Therapy. *N. Engl J Med*. 2017;377:2167–79.
- Strobl J, Pandey RV, Krausgruber T, Bayer N, Kleissl L, Reininger B, et al. Long-term skin-resident memory T cells proliferate in situ and are involved in human graft-versus-host disease. *Sci Transl Med*. 2020;12:eabb7028.
- Strong Rodrigues K, Oliveira-Ribeiro C, de Abreu Fiuza Gomes S, Knobler R. Cutaneous Graft-Versus-Host Disease: Diagnosis and Treatment. *Am J Clin Dermatol*. 2018;19:33–50.
- Mathewson N, Reddy P. The Microbiome and Graft Versus Host Disease. *Curr Stem Cell Rep*. 2015;1:39–47.
- Taur Y, Jenq RR, Perales MA, Littmann ER, Morjaria S, Ling L, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood*. 2014;124:1174–82.
- Peled JU, Gomes ALC, Devlin SM, Littmann ER, Taur Y, Sung AD, et al. Microbiota as Predictor of Mortality in Allogeneic Hematopoietic-Cell Transplantation. *N. Engl J Med*. 2020;382:822–34.
- Byrd AL, Belkaid Y, Segre JA. The human skin microbiome. *Nat Rev Microbiol*. 2018;16:143–55.
- Nakatsuji T, Hata TR, Tong Y, Cheng JY, Shafiq F, Butcher AM, et al. Development of a human skin commensal microbe for bacteriotherapy of atopic dermatitis and use in a phase 1 randomized clinical trial. *Nat Med*. 2021;27:700–9.
- Glucksberg H, Storb R, Fefer A, Buckner CD, Neiman PE, Clift RA, et al. Clinical manifestations of graft-versus-host disease in human recipients of marrow from HL-A-matched sibling donors. *Transplantation*. 1974;18:295–304.
- Filipovich AH, Weisdorf D, Pavletic S, Socie G, Wingard JR, Lee SJ, et al. National Institutes of Health consensus development project on criteria for clinical trials in chronic graft-versus-host disease: I. Diagnosis and staging working group report. *Biol Blood Marrow Transpl*. 2005;11:945–56.
- Herlemann DP, Labrenz M, Jürgens K, Bertilsson S, Waniek JJ, Andersson AF. Transitions in bacterial communities along the 2000 km salinity gradient of the Baltic Sea. *Isme j*. 2011;5:1571–9.
- Pjevac P, Hausmann B, Schwarz J, Kohl G, Herbold CW, Loy A, et al. An Economical and Flexible Dual Barcoding, Two-Step PCR Approach for Highly Multiplexed Amplicon Sequencing. *Front Microbiol*. 2021;12:669776.
- Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJ, Holmes SP. DADA2: High-resolution sample inference from Illumina amplicon data. *Nat Methods*. 2016;13:581–3.
- Callahan BJ, Sankaran K, Fukuyama JA, McMurdie PJ, Holmes SP. Bioconductor Workflow for Microbiome Data Analysis: from raw reads to community analyses. *F1000Res*. 2016;5:1492.
- Pruesse E, Peplies J, Glöckner FO. SINA: accurate high-throughput multiple sequence alignment of ribosomal RNA genes. *Bioinformatics*. 2012;28:1823–9.
- Quast C, Pruesse E, Yilmaz P, Gerken J, Schweer T, Yarza P, et al. The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. *Nucleic Acids Res*. 2013;41:D590–6.
- Kasper S, Andersen RHK, Søren M, Karst, Mads Albertsen. mpvis2: an R package to analyse and visualise 16S rRNA amplicon data. *bioRxiv*. 2019. <https://github.com/MadsAlbertsen/ampvis2>.
- Jari Oksanen FGB, Michael F, Roeland K, Pierre L, Dan McGlinn, Peter R, et al. *Vegan: Community Ecology Package*. R Package Version 2.5-7. 2020. <http://CRAN.R-project.org/package=vegan>.
- Amann R, Fuchs BM. Single-cell identification in microbial communities by improved fluorescence in situ hybridization techniques. *Nat Rev Micro*. 2008;6:339–48.
- Trebesius K, Leitritz L, Adler K, Schubert S, Autenrieth IB, Heesemann J. Culture independent and rapid identification of bacterial pathogens in necrotising fasciitis and streptococcal toxic shock syndrome by fluorescence in situ hybridisation. *Med Microbiol Immunol*. 2000;188:169–75.
- Daims H, Lückner S, Wagner M. daime, a novel image analysis program for microbial ecology and biofilm research. *Environ Microbiol*. 2006;8:200–13.
- Picelli S, Faridani OR, Björklund ÅK, Winberg G, Sagasser S, Sandberg R. Full-length RNA-seq from single cells using Smart-seq2. *Nat Protoc*. 2014;9:171–81.
- Gómez-Sánchez D, Schlötterer C. ReadTools: A universal toolkit for handling sequence data from different sequencing platforms. *Mol Ecol Resour*. 2018;18:676–80.
- Dobin A, Davis CA, Schlesinger F, Drenkow J, Zaleski C, Jha S, et al. STAR: ultrafast universal RNA-seq aligner. *Bioinformatics*. 2013;29:15–21.
- Li H, Handsaker B, Wysoker A, Fennell T, Ruan J, Homer N, et al. The Sequence Alignment/Map format and SAMtools. *Bioinformatics*. 2009;25:2078–9.
- Liao Y, Smyth GK, Shi W. featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics*. 2014;30:923–30.
- Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol*. 2014;15:550.
- Ritchie ME, Phipson B, Wu D, Hu Y, Law CW, Shi W, et al. limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res*. 2015;43:e47.
- Brüggen M-C, Klein I, Greinix H, Bauer W, Kuzmina Z, Rabitsch W, et al. Diverse T-cell responses characterize the different manifestations of cutaneous graft-versus-host disease. *Blood*. 2014;123:290–9.
- Martin M. Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnetjournal*. 2011;17:10–2.
- Bankevich A, Nurk S, Antipov D, Gurevich AA, Dvorkin M, Kulikov AS, et al. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. *J Comput Biol*. 2012;19:455–77.
- Parks DH, Imelfort M, Skennerton CT, Hugenholtz P, Tyson GW. CheckM: assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes. *Genome Res*. 2015;25:1043–55.
- Jain C, Rodriguez-R LM, Phillippy AM, Konstantinidis KT, Aluru S. High throughput ANI analysis of 90K prokaryotic genomes reveals clear species boundaries. *Nat Commun*. 2018;9:5114.
- Seemann T. Prokka: rapid prokaryotic genome annotation. *Bioinformatics*. 2014;30:2068–9.
- Lechner M, Findeiß S, Steiner L, Marz M, Stadler PF, Prohaska SJ. Proteinortho: Detection of (Co-)orthologs in large-scale analysis. *BMC Bioinforma*. 2011;12:124.
- Casson CN, Yu J, Reyes VM, Taschuk FO, Yadav A, Copenhagen AM, et al. Human caspase-4 mediates noncanonical inflammasome activation against gram-negative bacterial pathogens. *Proc Natl Acad Sci*. 2015;112:6688.
- Alloaati A, Kotsias F, Pauwels A-M, Carpiér J-M, Jouve M, Timmerman E, et al. Toll-like Receptor 4 Engagement on Dendritic Cells Restrains Phago-Lysosome Fusion and Promotes Cross-Presentation of Antigens. *Immunity*. 2015;43:1087–100.
- Buschow SI, Lasonder E, Szklarczyk R, Oud MM, de Vries IJM, Figdor CG. Unraveling the human dendritic cell phagosome proteome by organellar enrichment ranking. *J Proteom*. 2012;75:1547–62.
- Li S, Paulsson KM, Chen S, Sjögren HO, Wang P. Tapasin is required for efficient peptide binding to transporter associated with antigen processing. *J Biol Chem*. 2000;275:1581–6.
- Montoya M, Schiavoni G, Mattei F, Gresser I, Belardelli F, Borrow P, et al. Type I interferons produced by dendritic cells promote their phenotypic and functional activation. *Blood*. 2002;99:3263–71.
- Simmons DP, Wearsch PA, Canaday DH, Meyerson HJ, Liu YC, Wang Y, et al. Type I IFN Drives a Distinctive Dendritic Cell Maturation Phenotype That Allows Continued Class II MHC Synthesis and Antigen Processing. *J Immunol*. 2012;188:3116–26.
- Gérard A, Patino-Lopez G, Beemiller P, Nambiar R, Ben-Aissa K, Liu Y, et al. Detection of rare antigen-presenting cells through T cell-intrinsic meandering motility, mediated by Myo1g. *Cell*. 2014;158:492–505.
- Efremova M, Vento-Tormo M, Teichmann SA, Vento-Tormo R. CellPhoneDB: inferring cell-cell communication from combined expression of multi-subunit ligand-receptor complexes. *Nat Protoc*. 2020;15:1484–506.
- Golob JL, Pergam SA, Srinivasan S, Fiedler TL, Liu C, Garcia K, et al. Stool Microbiota at Neutrophil Recovery Is Predictive for Severe Acute Graft vs Host Disease After Hematopoietic Cell Transplantation. *Clin Infect Dis*. 2017;65:1984–91.
- Spindelboeck W, Schulz E, Uhl B, Kashofer K, Aigelsreiter A, Zinke-Cervenka W, et al. Repeated fecal microbiota transplantations attenuate diarrhea and lead to sustained changes in the fecal microbiota in acute, refractory gastrointestinal graft-versus-host-disease. *Haematologica*. 2017;102:e210–3.
- van Lier YF, Davids M, Haverkate NJE, de Groot PF, Donker ML, Meijer E, et al. Donor fecal microbiota transplantation ameliorates intestinal graft-versus-host disease in allogeneic hematopoietic cell transplant recipients. *Sci Transl Med*. 2020;12:eaa8926.

51. Holler E, Butzhammer P, Schmid K, Hundsrucker C, Koestler J, Peter K, et al. Metagenomic analysis of the stool microbiome in patients receiving allogeneic stem cell transplantation: loss of diversity is associated with use of systemic antibiotics and more pronounced in gastrointestinal graft-versus-host disease. *Biol Blood Marrow Transpl.* 2014;20:640–5.
52. Biagi E, Zama D, Nastasi C, Consolandi C, Fiori J, Rampelli S, et al. Gut microbiota trajectory in pediatric patients undergoing hematopoietic SCT. *Bone Marrow Transpl.* 2015;50:992–8.
53. Jenq RR, Taur Y, Devlin SM, Ponce DM, Goldberg JD, Ahr KF, et al. Intestinal Blautia Is Associated with Reduced Death from Graft-versus-Host Disease. *Biol Blood Marrow Transpl.* 2015;21:1373–83.
54. Gu Y, Sun J, Li K, Wu X, Zhang J. Alteration in the Skin Microbiome in Cutaneous Graft Versus Host Disease. *Acta Derm Venereol.* 2021;101:adv00374.
55. Ghimire S, Weber D, Mavin E, Wang Xn, Dickinson AM, Holler E. Pathophysiology of GvHD and Other HSCT-Related Major Complications. *Front Immunol.* 2017;8:79.
56. Lei YM, Sepulveda M, Chen L, Wang Y, Pirozzolo I, Theriault B, et al. Skin-restricted commensal colonization accelerates skin graft rejection. *JCI Insight.* 2019;5:e127569.
57. Hill GR, Crawford JM, Cooke KR, Brinson YS, Pan L, Ferrara JLM. Total Body Irradiation and Acute Graft-Versus-Host Disease: The Role of Gastrointestinal Damage and Inflammatory Cytokines. *Blood.* 1997;90:3204–13.
58. Nestel FP, Price KS, Seemayer TA, Lapp WS. Macrophage priming and lipopolysaccharide-triggered release of tumor necrosis factor alpha during graft-versus-host disease. *J Exp Med.* 1992;175:405–13.
59. Penack O, Smith OM, Cunningham-Bussell A, Liu X, Rao U, Yim N, et al. NOD2 regulates hematopoietic cell function during graft-versus-host disease. *J Exp Med.* 2009;206:2101–10.
60. Heimesaat MM, Nogai A, Bereswill S, Plickert R, Fischer A, Loddenkemper C, et al. MyD88/TLR9 mediated immunopathology and gut microbiota dynamics in a novel murine model of intestinal graft-versus-host disease. *Gut.* 2010;59:1079–87.
61. Williams MR, Cau L, Wang Y, Kaul D, Sanford JA, Zaramela LS, et al. Interplay of Staphylococcal and Host Proteases Promotes Skin Barrier Disruption in Netherton Syndrome. *Cell Rep.* 2020;30:2923–33.e7.

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AUTHOR CONTRIBUTIONS

Conceptualization, GS; Methodology, NB; Software, RN and MLW; Formal Analysis, NB, BH, RVP, FD, PP, CK and RK; Investigation, NB, LMG, JS, AR, VB, LK, MN and LH; Resources, CS, PW and WR; Data Curation, NB, BH and LU; Writing—Original Draft, NB; Writing—Review and Editing, GS, SK; DB and AM; Visualization, NB; Supervision, GS, SK; DB and RCE; Funding Acquisition, GS, NB.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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